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## **Project Title: Import Data using Transform Maps (Spreadsheet) & Visual Analytics**

**Program:** TN Skills (Vetri Thiran Payirchi Thittam)

**Track:** ServiceNow Certified System Administrator (CSA)

**Team ID:** SWTID-2026-7960

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### **1. Project Overview & Objective**

The primary objective of this project is to implement an automated, configuration-driven data integration pipeline within the ServiceNow platform. In real-world enterprise environments, migrating external employee records from raw spreadsheets into production tables requires strict data integrity, schema mapping, and duplicate prevention mechanisms.

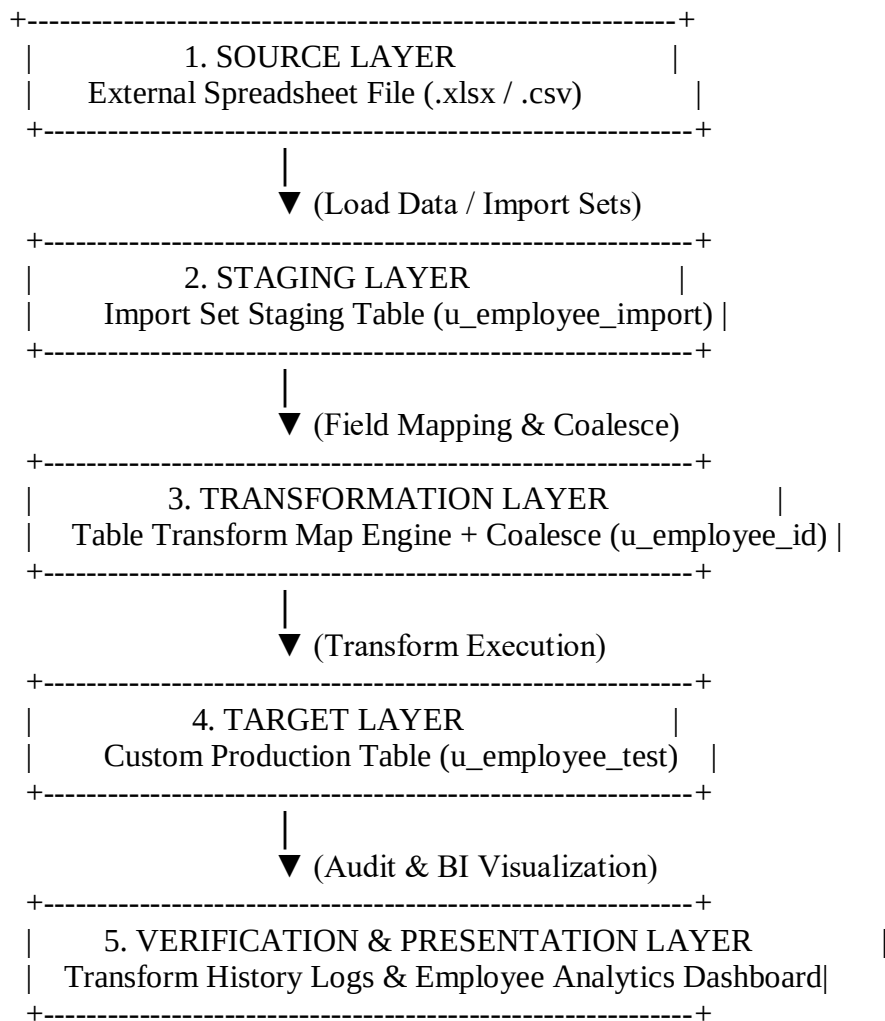
This project simulates an end-to-end data migration scenario where tabular employee records (containing Employee ID, Name, Email, Department, and Location) are loaded via Data Sources and Import Sets into a staging table (u\_employee\_import). Using Table Transform Maps, the data is accurately transformed and loaded into a custom target production table (u\_employee\_test) without data loss. Furthermore, to provide actionable business intelligence, an interactive **Employee Analytics Dashboard** was configured using the ServiceNow Platform Analytics Workspace, featuring multi-metric visualizations and dynamic reports.

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### **2. System Architecture & Data Flow**

The solution adopts a structured, multi-tier data flow architecture ensuring complete separation between raw source data, staging areas, transformation rules, target storage, and visual presentation:

1. **Source Layer:** External spreadsheet files (.csv / .xlsx) containing raw employee records.
2. **Staging Layer:** ServiceNow Import Set Staging Table (u\_employee\_import) that receives raw rows without altering production records.
3. **Transformation Layer:** Table Transform Map engine incorporating Auto Map Matching, Mapping Assist, and Coalesce validation rules.
4. **Target Layer:** Custom ServiceNow production table (u\_employee\_test) storing validated records.
5. **Presentation & Verification Layer:** Transform History audit logs and the Employee Analytics Dashboard.



#### Layer-by-Layer Architectural Description:

##### 1. Source Layer:

- *Role:* Acts as the external data origin supplying structured employee records (Employee ID, Name, Email, Department, Location) via Excel or CSV spreadsheet files.

##### 2. Staging Layer:

- *Role:* Ingests raw spreadsheet data into a temporary staging import set table (u\_employee\_import) without directly altering production records[cite: 1, 4].

##### 3. Transformation Layer:

- *Role:* Applies field mapping rules and duplicate prevention logic. Setting **Coalesce = true** on the u\_employee\_id field ensures that existing records are updated instead of generating duplicates[cite: 1, 4].

##### 4. Target Layer:

- *Role:* Securely stores the final transformed, validated, and clean employee records inside the custom target production table (u\_employee\_test)[cite: 1, 4].

##### 5. Verification & Presentation Layer:

- *Role:* Tracks execution logs via Transform History and provides real-time business intelligence using the centralized **Employee Analytics Dashboard** with interactive charts[cite: 1, 4].

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### 3. Project Workflow & Milestones

#### Milestone 1: Preparation & Schema Design

- **Activity 1.1:** Prepare structured spreadsheet templates containing mandatory attributes (Employee ID, Name, Email, Department, Location).
- **Activity 1.2:** Configure the custom target table (u\_employee\_test) within the ServiceNow Personal Developer Instance (PDI), defining appropriate string fields and column attributes.
- **Activity 1.3:** Establish user roles and access controls (RBAC) to ensure secure administrative data handling.

## Milestone 2: Import Set & Staging Configuration

- **Activity 2.1:** Utilize the ServiceNow **Load Data** module to ingest the raw spreadsheet into the newly generated staging import set table (u\_employee\_import).
- **Activity 2.2:** Verify successful staging table population and initiate the creation of a corresponding Table Transform Map (Sample Spreadsheet Import).

The screenshot shows the ServiceNow interface for configuring a Table Transform Map. The page title is "Table Transform Map - Sample Spreadsheet Import". The configuration is as follows:

- Name:** Sample Spreadsheet Import
- Source table:** Employee Import [u\_employee\_im...]
- Active:** ☒
- Run business rules:** ☒
- Enforce mandatory fields:** No
- Copy empty fields:** ☐
- Create new record on empty coalesce fields:** ☐
- Application:** Global
- Created:** 2026-09-28 06:32:00
- Target table:** Employee Test [u\_employee\_test]
- Order:** 100
- Run script:** ☐

Below the configuration fields, there are buttons for "Copy", "Update", and "Delete". A section titled "Related Links" contains the following links:

- [Auto Map Matching Fields](#)
- [Mapping Assist](#)
- [Validate Coalesce Fields](#)
- [Transform](#)
- [Index Coalesce Fields](#)
- [Run Point Scan](#)

At the bottom, there is a summary bar showing "Field Maps (5)", "Transform Scripts", and "Unused coalesce fields for empty values (1)".

- **Transform Map Name:** Sample Spreadsheet Import
- **Source Table:** Employee Import (u\_employee\_import)
- **Target Table:** Employee Test (u\_employee\_test)
- **Application:** Global
- **Order:** 100
- **Run Business Rules:** Active (true)

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### Milestone 3: Transform Map & Coalesce Validation

- **Activity 3.1:** Define source-to-target field mappings using Mapping Assist (e.g., mapping u\_employee\_id to u\_employee\_id, u\_name to u\_employee\_name, etc.).
- **Activity 3.2:** Implement duplicate prevention by setting **Coalesce = true** on the u\_employee\_id primary matching key. This ensures that repeated or updated imports modify existing records rather than generating redundant duplicate entries.

| Source Field  | Target Field    | Coalesce    |
|---------------|-----------------|-------------|
| u_employee_id | u_employee_id   | <b>true</b> |
| u_name        | u_employee_name | false       |
| u_email       | u_email         | false       |
| u_department  | u_department    | false       |
| u_location    | u_location      | false       |

dev267931.service-now.com/now/nav/ui/classic/params/target/sys\_transform\_map.do%3Fsys\_id%3D6d3559aec3a74310f69bb11ed40131cf%26sysparm\_recor...

serviceNowAllFavoritesHistoryTable Transform Map - Sample Sprea...Search

Table Transform MapSample Spreadsheet Import

[Auto Map Matching Fields](#)[Mapping Assist](#)[Validate Coalesce Fields](#)[Transform](#)[Index Coalesce Fields](#)[Run Point Scan](#)

Field Maps (5)Transform ScriptsUnused coalesce fields for empty values (1)

Source field

Target field

Coalesce

☐

u\_email

u\_email

false

☐

u\_name

u\_employee\_name

false

☐

u\_location

u\_location

false

☐

u\_department

u\_department

false

☐

u\_employee\_id

u\_employee\_id

true

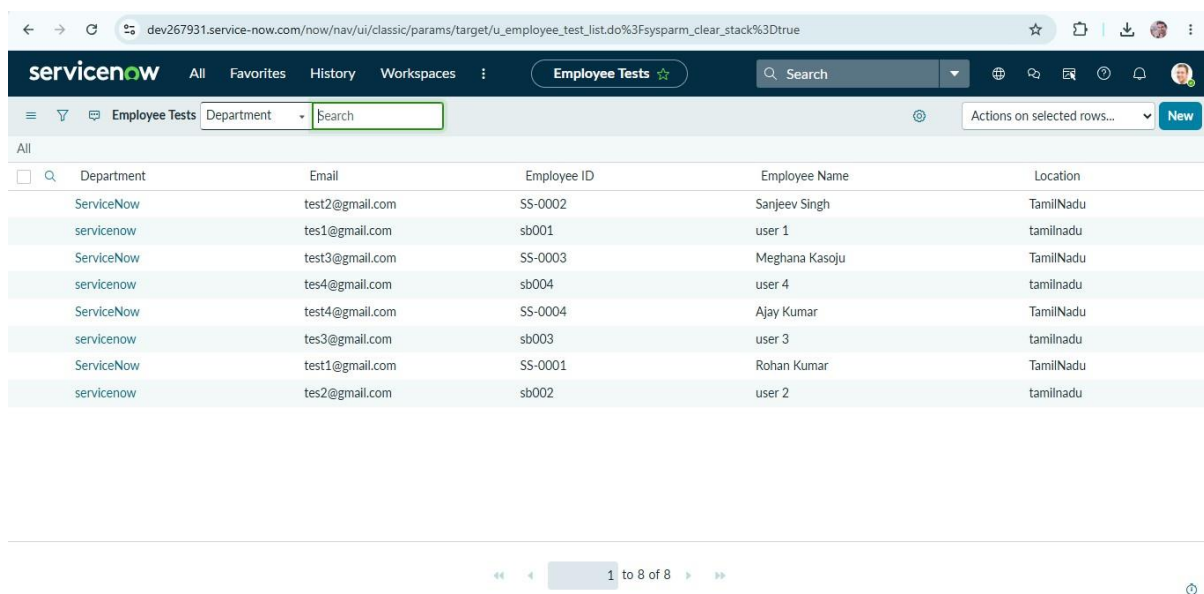
1 to 5 of 5

## 2. Milestone 4: Data Validation & Table Records List

**Description:** After running the transformation process, the target table was verified to confirm that all records were imported accurately with their corresponding attributes populated.

### Milestone 4: Execution & Data Validation

- **Activity 4.1:** Execute the transform operation and review the **Transform History** logs to confirm successful processing outcomes (Inserted / Updated / Skipped counts).
- **Activity 4.2:** Validate target table (u\_employee\_test) records to ensure complete data integrity across all 8 processed entries.



The screenshot shows a ServiceNow interface for the 'Employee Tests' table. The table has columns for Department, Email, Employee ID, Employee Name, and Location. There are 8 records listed. The interface includes a search bar, a filter dropdown, and a 'New' button. The bottom of the table shows a pagination bar indicating '1 to 8 of 8' records.

| Department | Email           | Employee ID | Employee Name  | Location  |
|------------|-----------------|-------------|----------------|-----------|
| ServiceNow | test2@gmail.com | SS-0002     | Sanjeev Singh  | TamilNadu |
| servicenow | tes1@gmail.com  | sb001       | user 1         | tamilnadu |
| ServiceNow | test3@gmail.com | SS-0003     | Meghana Kasoju | TamilNadu |
| servicenow | tes4@gmail.com  | sb004       | user 4         | tamilnadu |
| ServiceNow | test4@gmail.com | SS-0004     | Ajay Kumar     | TamilNadu |
| servicenow | tes3@gmail.com  | sb003       | user 3         | tamilnadu |
| ServiceNow | test1@gmail.com | SS-0001     | Rohan Kumar    | TamilNadu |
| servicenow | tes2@gmail.com  | sb002       | user 2         | tamilnadu |

- **Total Records Successfully Processed: 8 records**

- **Attributes Verified:** Employee ID, Employee Name, Email, Department, and Location.

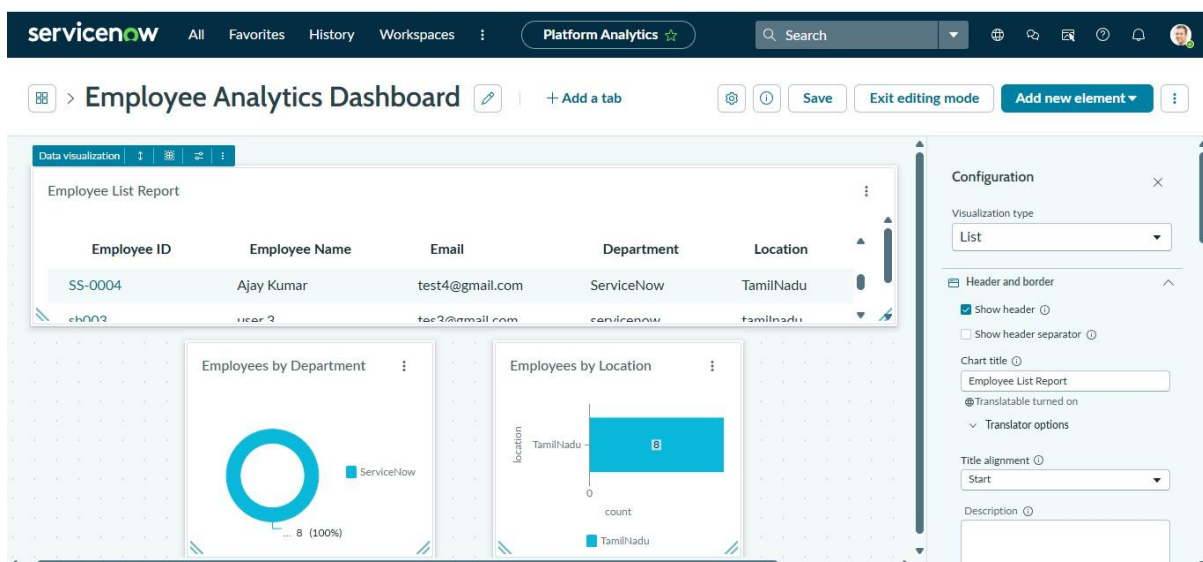
### 3. Milestone 5: Visual Analytics & Dashboard Creation

**Description:** Using ServiceNow Platform Analytics Workspace, an **Employee Analytics Dashboard** was created to deliver real-time visual metrics from the imported dataset.

□ **Activity 5.1:** Access the ServiceNow Platform Analytics Workspace to design operational reports based on the imported dataset.

□ **Activity 5.2:** Configure and consolidate the reports into a centralized **Employee Analytics Dashboard**:

- **Employee List Report (Data Table View):** Displays the complete listing of active employee records.
- **Employees by Department (Donut Chart):** Visualizes categorical distribution grouped by department.
- **Employees by Location (Horizontal Bar Chart):** Represents headcounts categorized across distinct regional locations.



- **Employee List Report (Data Table View):** Displays the complete listing of active employee records.

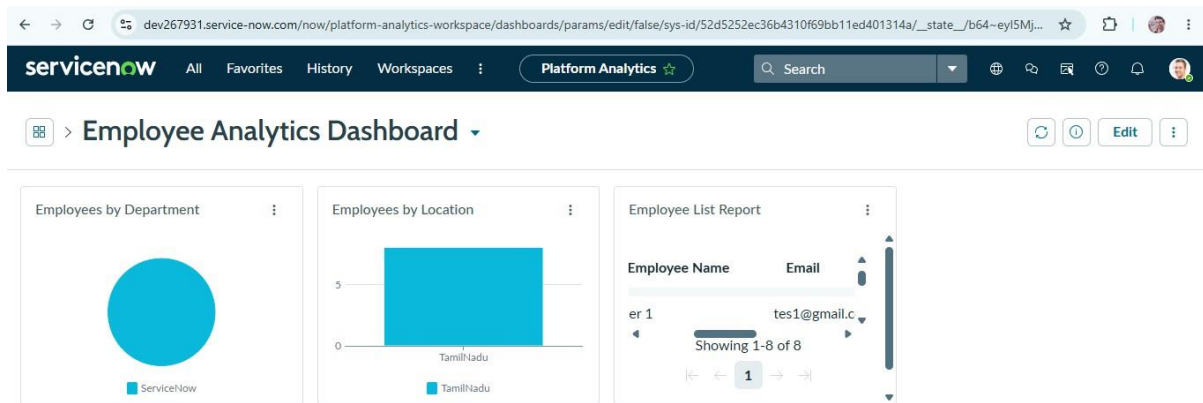
- **Employees by Department (Donut Chart):** Visualizes the categorical distribution of employees grouped by department.
- **Employees by Location (Horizontal Bar Chart):** Represents headcounts categorized across distinct locations.

### **Conclusion & Future Enhancements**

The implementation successfully demonstrates external data ingestion, automated schema mapping, duplicate prevention through Coalesce configuration, and operational business intelligence reporting via dynamic ServiceNow dashboards.

#### **Future Roadmap:**

- **Automated Scheduled Imports:** Integrating SFTP or REST API data feeds to eliminate manual spreadsheet uploads.
- **Advanced Error Alerting:** Setting up automated notification workflows when transform exceptions or missing mandatory fields occur.
- **AI-driven Data Anomaly Detection:** Implementing predictive validation checks before schema transformation.



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#### 4. Conclusion

The implementation successfully demonstrates external data ingestion, automated schema mapping, duplicate prevention through coalesce configuration, and operational business intelligence reporting via dynamic ServiceNow dashboards.